

UNIT 11 Survival in an emergency

On board the Marina, like on most merchant vessels, there are clear instructions for the crew on how to act during an emergency. Every crew member has their own specific duties, and when the alarm goes everyone immediately rushes to their station to carry out their tasks. All the rescue equipment on board has to fulfil the rules and regulations given in the SOLAS, The Convention for Safety of Life at Sea.

Let us follow a boat drill on board the MS Marina.

The following is heard on the loudspeaker...

This is your Captain speaking. All officers report to the bridge. All crew members to assembly stations!

Leave the engine-room, the superstructure and your cabins immediately!

Take lifejackets and emergency equipment with you according to the muster list.

Do not go to the lifeboat stations before ordered or enter the life-rafts and lifeboats - the order to enter will be given by the officers.

EVACUATION AND ROLL CALL

A: Evacuate all decks, rooms and spaces and report!

B: All decks, rooms and spaces evacuated.

A: Assembly station 1. This is Bridge. Report on number of crew at assembly station!

B: Bridge. This is Assembly station 1. Number of crew members at station is not yet complete.

A: Assembly station 1. This is Bridge. Search for missing crew members and report.

B: Bridge. Assembly station 1. Missing crew member recovered.

ABANDONING VESSEL

A: Lifeboatmen! Check the outfit of crew members at assembly stations and report!

B: Outfit of crew is complete.

A: Put on warm clothing - long trousers, head covering and report. Crew! Follow the lifeboatmen to the lifeboat stations and life-raft stations on the embarkation deck.

A: Swing out number 1 and 2 lifeboats and report.

B: Number 1 and 2 lifeboats swung out.

A: Lower number 1 and 2 lifeboats alongside the embarkation deck. Enter the lifeboats via the deck. Hold on to your seats while launching.

A: Throw overboard life-raft and report!

B: Life-raft thrown overboard.

Enter the life-rafts via the slides, ladder, nets and manropes.

Jump onto the life-raft alongside the vessel.

Clear the entrance of the life-raft.

Jump into the water and enter the life-rafts.

Assist helpless people.

Sit down in the life-rafts immediately.

- Report on number of injured persons!
- No persons injured.
- Let go sea anchor and report!
- Sea anchor is let go.
- Fire rockets for identification.
- Connect the life-boats and rafts with lines and report!
- Life-rafts and lifeboats connected.

Now let's learn some more about survival in an emergency. The following advice can be found on the Survival Card published by The Nautical Institute in Great Britain:

WHEN YOU HEAR AN EMERGENCY SIGNAL

Go to your station.

If possible, collect warm clothes, waterproof clothing and a lifejacket from their stowage.

Have a good drink of water.

IF YOU HAVE TO ABANDON SHIP

Put on waterproof clothes, rescue suits and lifejacket.

Assist in loading extra water, provisions (not protein food) and blankets.

Release lifeboat or cut life-raft painter.

Secure painters, launch lifeboats/rafts and board safe if possible.

Assist clearing the ship's side and danger area, stream the sea anchor.

Avoid unnecessary swimming if you have to enter the water.



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SURVIVAL AFTER SHIPWRECK / GENERAL

Help protect yourself from the environment:

- take anti-seasickness pills
- inflate floor and use doors to regulate conditions in life-rafts
- in open lifeboats, erect canopy. Keep as warm and dry as possible.
- follow the instructions in the survival equipment.

POST COMMAND

The senior fit survivor must take command and decide on:

- the distribution of survivors (full rafts in cold climates, as few as possible in hot climates)
- a congregation of rafts and boats to make a better target for air search
- navigation
- fair rationing of food supplies
- distribution of duties, lookouts and routine

FIRST PRIORITY: PROTECTION

You should:

- bail out the boat/raft, inflate life-raft floor, rig canopy (boats), close entrances
- keep the raft/ boat dry, prevent reswamping, keep head to wind and cast sea anchor
- check for leaks and make repairs, collect useful flotsam
- apply first aid
- treat for asphyxia / breathing difficulties; (clear airways, commence mouth-to-mouth)
- stop bleeding (apply pad and bandages)
- treat for shock (keep warm, encourage, relieve pain)
- treat fractures
- prevent sea sickness (everyone should take a pill if available)
- keep clothing on. Wet clothing is much better than no clothes. Insulate yourself from weather and water as well as you can.

SECOND PRIORITY: LOCATION

You can do the following to assist location of the raft:



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- stay where you are. Normally the search will start in the vicinity of the wreck
- prepare your signalling devices, and mount a lookout. Put the emergency transmitter to use.
- read survival literature in the life-raft survival pack.

THIRD PRIORITY: WATER

You should start a strict rationing at once.

- normally drink nothing for the first 24 hours
- then distribute not less than one pint per person per day and be prepared to catch any rain water
- do not drink any sea water or alcohol. Injured people need extra water.

FOURTH PRIORITY: FOOD

You don't need food for long periods so:

- unless there is a plenty of water, only eat carbohydrates such as sweets, glucose, etc.
- do not eat protein, it will dehydrate you; cigarettes also dehydrate you so do not smoke.

IN COLD CLIMATES

Remember to:

- protect face, ears and hands from frostbite, keep each other warm, watch another for white patches. Do not massage frostbite.
- keep feet as dry as possible. Move the fingers and toes, move at ankles and knees.
- keep the blood circulating.
- put feet up for at least five minutes every hour.
- keep weather cover closed except for small opening to ensure ventilation.

Extra text: FIGHTING FIRE ON BOARD

Of all the disasters that can happen to a vessel at sea, fire is one of the most devastating. Marine fire incidents are very destructive and have been responsible for the loss of many lives. Ship fires are second only to shipwrecks when calculating casualties and total loss.

It can often be a single careless act that endangers the lives of all the crew. That is why every member of the crew has a part to play in preventing fires. This applies not only to fire-fighting methods. There are a number of precautions crew members can take on a day-to-day basis to make the ship a safer place and prevent the break-out of fires. The main danger spots are machinery, galleys, accommodation and cargo spaces.

What can we do to make the ship a safer place?

- Careless smoking is a common cause of fires on ships. Always put out cigarettes completely. Smoking and non-smoking areas must be clearly defined. Comply with no-smoking signs. No one should be allowed to smoke e.g. in a ship's hold or the engine-room.
- Cleanliness is important in all parts of the ship. Do not allow flammable waste to pile up or accumulate. This is a special risk in the engine room, where heated parts of the engine may give off sparks igniting surrounding materials and where dirt, grease, rags and cotton waste might easily self-ignite, if they come in contact with each other.
- Keep oil away from hot surfaces which might ignite it - check that all feed pipes and joints are in good condition. Be careful when pumping oil to ready- use tanks - avoid spillage or overflow which could lead to fire.
- All electrical equipment from light bulbs to radiators and heaters produce heat, which may cause fire without the need for a spark. Switch off electrical equipment when not in use, whenever possible. Be particularly careful in galleys and cargo holds. Do not allow light clusters or single lights to be buried in the cargo - always check after loading.
- Always keep fire doors closed to restrict the spread of flames and smoke.
- Education is, as always, the greatest aid to safety. During fire drills on board everyone must learn where fire-fighting equipment and lifesaving appliances are kept and how they work.
- Station bills. Each vessel should have a Fire Duty Station Bill posted in a visible place and every crew member should be clear about his role and duties in a fire emergency.

Equipment

The current practice is to arrange special fire fighting courses, during which crew members are taught different kinds of extinguishing agents and fire-fighting methods



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and the use of various fire extinguishers such as water, sprinklers, the use of foam, CO2 gas, powder and fire smothering blankets. The students are trained in using fire-axes, fire hoses and nozzles. Smoke and toxic substances in the air make it necessary to use breathing apparatus. Gas and dust masks, protective clothing like fire protection suits, helmets, goggles, gloves, footwear and fire-proof lifelines facilitate the work of fire-fighters, but much of the equipment requires that the wearer has received thorough training in its use.

Needless to say it is vitally important to make sure that all fire fighting equipment on board measures up to SOLAS standards. Fire mains should not be leaking, hydrants should not fail to supply pressurised water to the fire pumps, including the emergency pump systems, which must be operable under all conditions, hoses, sprinklers and breathing apparatus must be in good condition.